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Transformation of Vietnamese Kaolin into Zeolite Y–Templated Mesoporous Aluminosilicate for Efficient Catalysis in Polyolefin–Hydrocarbon Co-Pyrolysis

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Abstract: *This study presents the first successful transformation of Vietnamese kaolin into a hierarchical mesostructured aluminosilicate catalyst, MAS(Y), via a sustainable two-step synthesis. Unlike conventional methods relying on high-purity chemicals, this approach employs thermally activated local kaolin and glass water to generate reactive aluminosilicate species, eliminating the need for petrochemical-derived precursors. The MAS(Y) material integrates ordered mesopores templated by CTABr with embedded zeolite Y domains, as confirmed by SAXS, TEM, and XRD analyses. It exhibits a high specific surface area (971 m²/g), narrow pore size distribution (~3.2 nm), and thermal stability exceeding 900 °C. The NH_3-TPD profile demonstrates the coexistence of weak (207.8 °C) and strong acid sites (447.0 °C, 510.4 °C), attributed to surface silanol groups and framework Al^{3+} within zeolitic motifs. When applied in the catalytic co-pyrolysis of waste polypropylene (PP) with vacuum gas oil (VGO), MAS(Y) achieved liquid product yields exceeding 80%. Notably, the C_5–C_{12} and C_{13}–C_{18} hydrocarbon fractions peaked at PP/VGO ratios of 3:7 and 5:5, respectively. This work establishes a cost-effective and environmentally compatible pathway for upcycling abundant Vietnamese kaolin into high-performance acid catalysts, offering strong prospects for plastic waste valorization and advanced refinery applications.*

Keywords: *Vietnamese kaolin, Mesoporous aluminosilicate, Zeolite Y, catalytic co-pyrolysis, plastic waste valorization*

1. Introduction

Kaolin is a geologically abundant and industrially valuable clay material, primarily composed of the mineral kaolinite, with minor contributions from halloysite, illite, muscovite, smectite, microcrystalline quartz, feldspar, and gibbsite (Fabbri et al., 2013; Irfan Khan et al., 2017). Vietnam holds substantial kaolin reserves - second only to China and India in the Asia-Pacific region—with an estimated 300 million tons distributed across 67 surveyed deposits and mining sites (Le.D.T et al, 2008).

Kaolinite ($Al_2Si_2O_5(OH)_4$) is a naturally occurring dioctahedral aluminosilicate with a 1:1 layered structure comprising silica tetrahedra (SiO_4) and alumina octahedra (AlO_6). It typically contains 46.54% SiO_2, 39.5% Al_2O_3, and 13.96% structural water (Irfan Khan et al., 2017; Khairy et al., 2018). The layers are held together through hydrogen bonding, with an interlayer spacing of approximately 7.1–7.2 Å (Hubadillah et al., 2018).

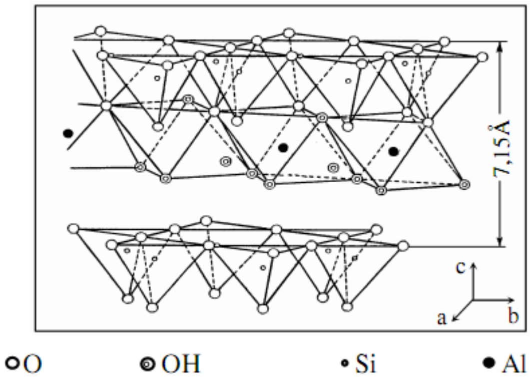

Fig. 1. Three-dimensional framework structure of kaolinite (Hubadillah et al., 2018)

In kaolinite, SiO_4 tetrahedra are oriented with their apexes toward AlO_6 octahedra, creating interlayer regions where O^{2-} and OH^- ions face each other. This asymmetry leads to hydrogen bonding between layers, imparting structural rigidity and thermal stability. Within the octahedral sheet, only two-thirds of sites are occupied by Al^{3+}, while the remaining are vacant. Hydroxyl groups present in this layer are progressively removed upon heating above 550 °C, initiating dehydroxylation (Khairy et al., 2018). Thermal treatment of kaolinite ($Al_2Si_2O_5(OH)_4$) yields amorphous metakaolinite ($Al_2Si_2O_7$) at 600–750 °C. Further heating leads to the formation of a spinel phase ($MgAl_2O_4$) around 950–985 °C and crystallization of mullite ($Al_2Si_2O_{13}$) above 1000 °C (Inocente et al., 2021; Khairy et al., 2018).

Kaolin is a versatile raw material widely applied in ceramics, paints, paper, cement, and chemical industries (Awugichew et al., 2025). In materials chemistry, it serves as a sustainable and cost-effective precursor for synthesizing catalysts and adsorbents (Yavuz and Saka, 2013). The shift toward greener chemical processes emphasizes not only the use of renewable raw materials but also environmentally benign synthesis methods (He et al., 2021; Magdy et al., 2017). Kaolin has been effectively used to produce zeolites and mesoporous aluminosilicates with low Si/Al ratios, including zeolite X (Feudjio et al., 2025; Jamil et al., 2011), Y (Gandhi et al., 2021; Krongkrachang et al., 2019), A (Ayele et al., 2015; Maia et al., 2019), MSU-S (Strzemiecka et al., 2012), and Al-MCM-41 (Chen et al., 2019). These materials offer high catalytic efficiency and adsorption capacity, making them valuable in catalysis, separation, and environmental applications (Feudjio et al., 2025; Gandhi et al., 2021).

Kaolinite exhibits low cation exchange capacity (CEC), limited adsorption, and negligible catalytic activity due to its 1:1 layered structure and small surface area (15–20 m^2/g) (Li et al., 2015). Its CEC typically ranges from 3–15 meq/100 g, reflecting modest surface charge and limited ion exchange potential (Yavuz and Saka, 2013). Unlike swelling clays, kaolinite's non-expandable structure restricts its use as a catalyst or support. These limitations sharply contrast with the superior ion exchange and adsorption properties of crystalline aluminosilicates such as zeolites. Accordingly, the conversion of kaolinite into zeolites or mesoporous materials is of both theoretical interest and practical relevance for catalysis and adsorption in industrial applications.

In Vietnam, kaolin deposits are widely distributed across northern and central provinces, yet their high-value utilization remains limited. The clay is often employed in low-end applications such as ceramics or fillers, despite its chemical composition—rich in SiO_2 and Al_2O_3—being favorable for advanced materials synthesis. Several studies have demonstrated the successful conversion of kaolin into crystalline zeolites, including NaA, NaY, and ZSM-5, by employing hydrothermal methods with appropriate structure-directing agents. These transformations typically involve the use of commercial silica and alumina precursors to supplement or replace the native framework-forming elements in kaolin.

However, the development of mesostructured aluminosilicates, particularly MAS(Y)-type materials, directly from kaolin remains relatively unexplored. Compared to microporous zeolites, mesoporous aluminosilicates offer larger pore diameters (2–10 nm), higher external surface area, and improved molecular diffusion—attributes that are highly desirable in applications such as catalysis, adsorption, and waste valorization. Most conventional synthesis routes for these materials rely on expensive silicon and aluminum sources (e.g., TEOS, aluminum isopropoxide) and require environmentally intensive template removal steps, which limits their scalability and sustainability.

At the same time, co-pyrolysis of polyolefin plastics with petroleum-derived residues like vacuum gas oil (VGO) is emerging as a strategy for converting waste into usable fuels and chemicals. However, this process demands catalysts that combine acidic functionality, thermal resistance, and mesoporosity to enable selective cracking and resist coke formation. Traditional zeolites such as Y or ZSM-5, while acidic and thermally stable, suffer from rapid deactivation due to their narrow micropore systems. Mesostructured catalysts based on kaolin-derived aluminosilicates could overcome these limitations while utilizing abundant local resources.

In this context, the present study investigates the synthesis of mesoporous aluminosilicate materials MAS(Y) via a two-step hydrothermal method using Vietnamese kaolin as the sole silica and alumina source. The work systematically explores synthesis conditions, structural evolution, and physicochemical properties of the resulting materials. Moreover, the catalytic activity of MAS(Y) is evaluated in the co-pyrolysis of polypropylene (PP) and VGO, with a focus on yield distribution and stability. By valorizing a readily available clay resource and avoiding commercial chemicals, this research aims to contribute to the development of sustainable, low-cost, and scalable pathways for mesoporous catalyst production, aligned with the goals of green chemistry and circular economy.

2. Materials and methods

2.1. Materials

Kaolin used in this study was sourced from a naturally occurring deposit processed by a mineral beneficiation facility located in Phu Tho Province, Vietnam. Glass water with a density of 1.36 g/mL was obtained from a local manufacturer in Vietnam. Waste polypropylene was collected from a local factory, and the VGO fraction was obtained from a refinery in Vietnam. Sodium hydroxide (NaOH, 98%) and hydrochloric acid (HCl, 37%), both of analytical grade, were procured from commercial suppliers based in Deajung, Korea. Cetyltrimethylammonium bromide (CTAB, ≥99%) was purchased from Sigma-Aldrich (USA) and used as received without further purification.

2.2. Activation of kaolin

The kaolin utilized for the synthesis of mesoporous zeolite-like material was obtained from a kaolin mining and processing facility in Phu Tho Province, Vietnam. This naturally occurring clay is predominantly composed of alumina (Al_2O_3) and silica (SiO_2).

To ensure the purity of the starting material, raw kaolin underwent a multi-step purification process to eliminate mineral and organic impurities, including quartz sand, calcite, pyrite, feldspar, and humic substances. The kaolin was dispersed in deionized water, thoroughly agitated, and subjected to successive decantation cycles. The suspension was subsequently filtered, and the recovered solid was oven-dried at 105 °C. The dried sample was then ground to a uniform particle size of 0.25 mm.

The purified kaolin was acid-leached using 4N hydrochloric acid at a solid-to-liquid ratio of 2:3. The suspension was stirred continuously for 6 h at 95 °C to facilitate the removal of exchangeable cations and soluble phases. After acid treatment, the solid was washed repeatedly with distilled water until no residual chloride ions were detected (as confirmed by $AgNO_3$ test), then dried at 105 °C. The resulting acid-treated kaolin (KA) was subsequently calcined in air at 650 °C for 3 h, using a controlled heating rate of 2 °C/min. This calcined product, referred to as activated kaolin (AK), served as the aluminosilicate source for the subsequent mesoporous material synthesis.

2.3. Conversion of Activated kaolin into mesoporous alumiluminosilicate material (MAS)

Step 1: Preparation of zeolite Y seed from activated kaolin

Activated kaolin (AK) was first dispersed in a sodium hydroxide (NaOH) solution and stirred thoroughly to promote dissolution and ensure uniform mixing. A glass water was subsequently introduced to adjust the Si/Al molar ratio, yielding a homogeneous gel with a characteristic milky-white appearance. The final gel composition was approximately $4Na_2O \cdot Al_2O_3 \cdot 6SiO_2 \cdot 160H_2O$.

Step 2: Assembly of mesoporous material structure from zeolite Y seed

The gel was aged under continuous stirring at ambient temperature and atmospheric pressure for 48 hours. Following this period, the gel, which contained Y-type zeolite precursors, was diluted with distilled water to a defined volume. A surfactant solution of cetyltrimethylammonium bromide (CTABr) was then added at a molar ratio of CTAB/(Si + Al) = 0.25. The pH was adjusted to 9 using concentrated hydrochloric acid (37%).

After additional stirring for 2 hours, the mixture was subjected to hydrothermal crystallization at 95 °C for 96 hours. The resulting solid was recovered by filtration and washed repeatedly with deionized water until the filtrate reached neutral pH. The solid was then dried at 105 °C, ground, and passed through a 0.15 mm sieve.

To eliminate the surfactant, the dried powder was calcined in air at 540 °C for 6 hours, using a heating rate of 2 °C/min. The final product, referred to as MAS(Y), was obtained as a fine, porous white powder suitable for further characterization.

2.4. The co-pyrolysis of PP waste and VGO fraction with MAS(Y) catalyst

The co-pyrolysis process of the VGO and waste PP mixture was carried out in a microflow fixed-bed reactor equipped with a programmed temperature control system (Fig 2). A predetermined ratio of PP and VGO was heated to 120 °C to form a homogeneous mixture. An amount of 0.4 g of catalyst was packed into the reactor tube and held in place by a layer of glass wool. When the temperature inside the reactor reached 500 °C, the feed mixture was injected into the reactor at a flow rate of 10 mL/min using N_2 flow as the carrier gas. The resulting liquid products were condensed and collected in a receiving flask.

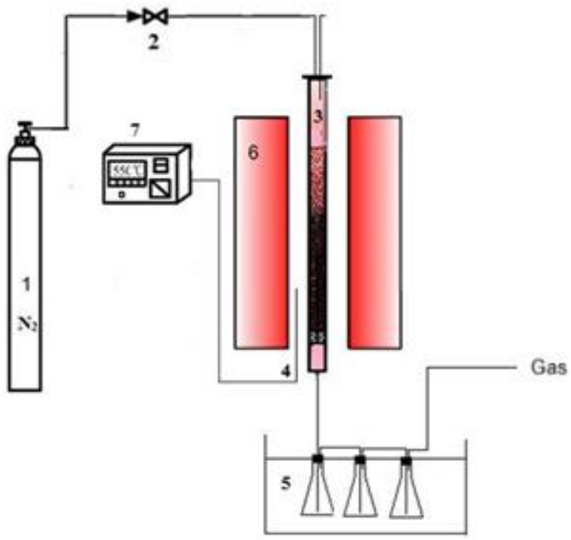

Fig. 2. Schematic diagram of the co-pyrolysis reaction system

1- Nitrogen cylinder; 2- Gas flow control valve; 3- Reactor tube; 4- Thermocouple; 5-cooled trap; 6- Heating furnace; 7- Temperature control unit

The yield of each fraction (Y) has been defined as the mass ratio of each particular fraction referred to the total mass fed:

$$Y(wt\%)=\frac{Mass_{fraction}}{Mass_{total}}\times 100$$

2.5. Material characterizations

The chemical composition of kaolin and the synthesized MASY material was analyzed by energy-dispersive X-ray spectroscopy (EDX) and Fourier-transform infrared spectroscopy (FTIR). Mesostructural features were characterized by small-angle X-ray scattering (SAXS) using a Bruker D8 Advance diffractometer (Cu Kα, λ = 1.5406 Å) equipped with a graphite monochromator, operated at 40 kV and 40 mA, over a 2θ range of 0.5°–10° with a scanning rate of 0.025°/s. Transmission electron microscopy (TEM) images were acquired on a JEOL 3010 microscope (80–300 kV), providing a spatial resolution of 0.17 nm (general) and 0.14 nm (lattice imaging) at magnifications of 50,000× to 200,000×.

Textural properties were assessed by nitrogen adsorption–desorption isotherms recorded at −196 °C on a Micromeritics ASAP 2060 system after degassing at 150 °C for 12 h. The specific surface area was calculated by the BET method; pore size distribution was derived from the BJH method (adsorption branch), and total pore volume was determined at P/Po = 0.99.

Thermal properties were examined using thermogravimetric analysis (TGA) on a SETARAM analyzer in nitrogen flow, from ambient to 1000 °C at a heating rate of 10 °C/min. Acidity was probed via temperature-programmed desorption of ammonia (NH_3-TPD) using an AutoChem II 2920 analyzer. Samples were pretreated in He at 200 °C for 1 h, saturated with NH_3 at 100 °C, purged in He for 1 h, and then desorbed from 100 °C to 610 °C at 10 °C/min.

For pyrolysis product analysis, the liquid fraction from PP/VGO co-pyrolysis was analyzed by GC–MS (Agilent, USA) using an HP-35 column (30 m × 0.25 mm × 0.25 µm). The oven was ramped from 40 °C to 280 °C at 10 °C/min and held for 5 min. Injector and transfer line temperatures were set at 250 °C; the injection was performed in splitless mode with 1 µL sample volume and a carrier gas flow of 1.5 mL/min.

3. Results and Discussions

3.1. Characteristics of MAS(Y) material

X-ray diffraction method:

X-ray diffraction (XRD) and small-angle X-ray scattering (SAXS) analyses were performed to elucidate the phase evolution during the thermal activation of kaolin and subsequent mesostructure formation (Fig 3).

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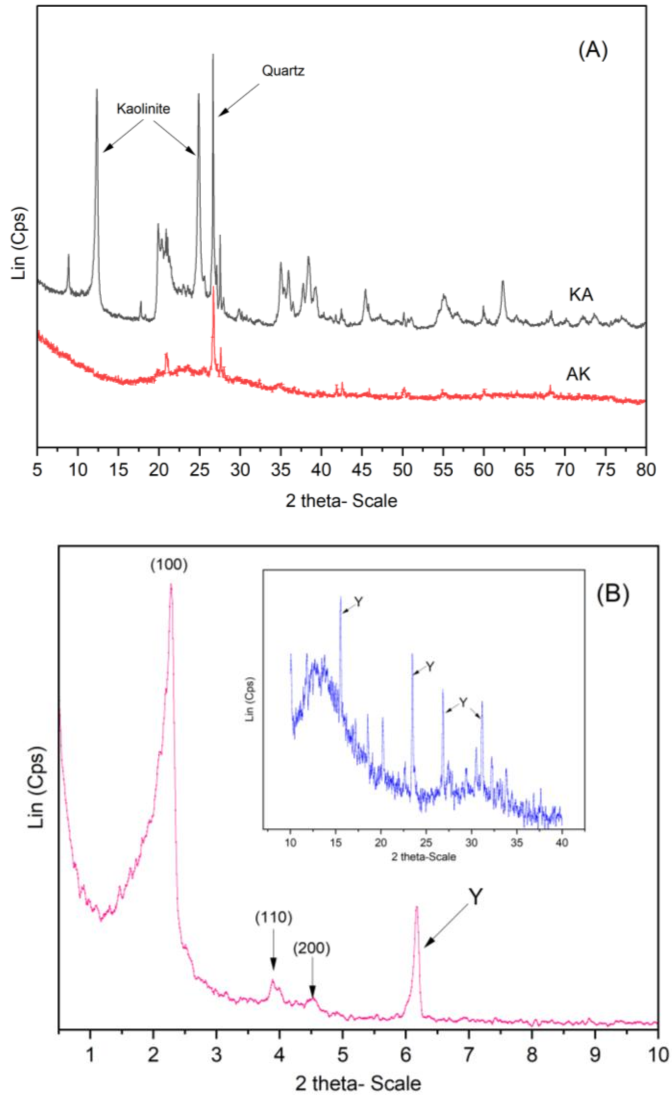

Fig. 3. The XRD patterns of raw kaolin (KA), activated kaolin (AK) and the SAXS of MAS(Y) (A) and the large X- ray diffraction of MAS(Y) (inset B)

The XRD pattern of raw kaolin (KA) (Fig 3A) exhibits intense reflections characteristic of crystalline kaolinite and quartz. The (001) and (002) planes of kaolinite are identified at $2\theta \approx 12.3°$ and $24.9°$, while quartz is indicated by a sharp peak at $2\theta \approx 26.6°$ (Gandhi et al., 2021; Lim et al., 2021)(D. Gandhi et al., 2021; W.-R. Lim et al., 2021), confirming the well-ordered layered structure of the parent material.

Thermal treatment at 650 °C results in complete disappearance of kaolinite peaks, alongside a marked reduction in quartz intensity. This transformation corresponds to dehydroxylation of kaolinite ($Al_2Si_2O_5(OH)_4 \rightarrow Al_2Si_2O_7 + 2H_2O$), leading to structural collapse of the octahedral sheets and formation of amorphous metakaolin (Y. He et al., 2021). The loss of long-range order is consistent with the transition to an X-ray amorphous phase (He et al., 2021). Metakaolin's structure, consisting of SiO_4 tetrahedra linked with AlO_4 tetrahedra, imparts high reactivity, enabling it to easily participate in chemical reactions such as those in the synthesis of zeolites and mesoporous aluminosilicates (Chen et al., 2019; Jamil et al., 2011).

Under alkaline hydrothermal conditions, the metakaolin framework reorganizes via condensation of TO_4 units (T = Si, Al), forming secondary building units (SBUs) that drive the nucleation of both zeolitic and mesostructured domains (Gandhi et al., 2021). The SAXS pattern of MAS(Y) (Fig 1B) shows three distinct reflections at $2\theta \approx 2.2°$, $3.8°$, and $4.6°$, indexed to the (100), (110), and (200) planes of a 2D hexagonal mesophase, confirming the formation of an ordered mesoporous architecture (Zhai et al., 2005). A weak diffraction signal at $2\theta \approx 6.2°$ further suggests the incipient formation of zeolite Y nanodomains. This is supported by the wide-angle XRD inset, which displays broad peaks at $2\theta \approx 16°$, $24°$, $28°$, and $31°$, consistent with the FAU-type zeolite Y framework (Gandhi et al., 2021; Liu et al., 2013). The absence of a sharp peak at $2\theta = 16°$ and the presence of a high background indicate that the zeolite nuclei formed from

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the activated kaolin have partially crystallized into zeolite crystals. Therefore, the MAS(Y) mesoporous material has walls that contain both zeolite Y structural units (zeolite seeds) and zeolite Y nanocrystals with low crystallinity (Krongkrachang et al., 2019; Liu et al., 2013).

The mesostructural order of MAS(Y) was quantitatively evaluated based on its SAXS pattern (Fig 3B). The diffraction peaks observed at $2\theta \approx 2.2°$, 3.8°, and 4.6° were assigned to the (100), (110), and (200) planes, respectively, which are characteristic of a two-dimensional (2D) hexagonal mesophase (p6mm symmetry) templated by CTABr surfactant micelles.

The interplanar spacing corresponding to the (100) reflection was calculated using Bragg's law:

$$d = \frac{\lambda}{2 \sin \theta}$$

Given that the (100) peak appears at $2\theta \approx 2.2°$, and using the Cu Kα radiation (λ = 1.5406 Å), the corresponding d-spacing is:

$$d_{100} = \frac{1.5406}{2\sin(1.1^{\circ})} \approx 4.02 \text{ nm}$$

From this, the unit cell parameter (a_0) of the 2D hexagonal lattice was determined using the relation:

$$a_0 = \frac{2}{\sqrt{3}} \cdot d_{100} \approx 4.64 \text{ nm}$$

Using this calculated a_0 value, the expected d-spacings for the (110) and (200) reflections were computed as:

$$d_{110} = \frac{a_0}{\sqrt{h^2+hk+k^2}} = \frac{a_0}{\sqrt{3}} \approx 2.68 \text{ nm}$$

$$d_{200} = \frac{a_0}{2} \approx 2.32 \text{ nm}$$

These values correspond to theoretical 2θ positions of approximately 3.8° and 4.6°, which match closely with the experimental SAXS data. This consistency among calculated and observed peak positions confirms the formation of a well-ordered 2D hexagonal mesostructure in MAS(Y).

Transmission electron microscopy (TEM):

Transmission electron microscopy (TEM) was employed to gain direct structural insight into the morphology and mesoscopic ordering of the MAS(Y) material (Fig 4).

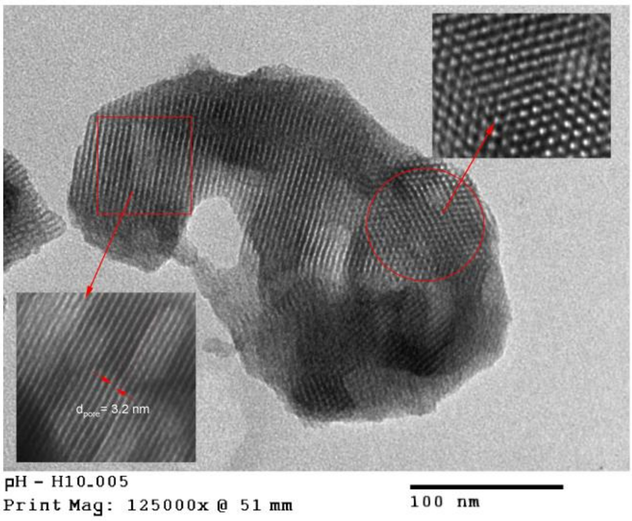

Fig. 4. TEM images of the MAS(Y) material

As presented in Fig. 4, the low-magnification TEM image reveals a uniform arrangement of parallel mesoporous channels aligned along a common axis. This structural motif is indicative of a two-dimensional hexagonal symmetry, reflecting a high degree of order throughout the material.

The enlarged region (left inset) displays distinct mesopore walls with an interchannel distance of approximately 3.2 nm. This spacing correlates closely with the (100) d-spacing observed in SAXS measurements ($2\theta \approx 2.2°$), confirming the internal consistency between direct imaging and scattering

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techniques. The data collectively affirm the formation of a well-organized mesoporous network derived from CTABr-directed assembly.

In addition, the high-resolution region (right inset) shows periodic lattice fringes corresponding to nanoscale crystalline domains embedded within the mesopore walls. These features are consistent with low-crystalline zeolite Y domains, as previously inferred from the wide-angle XRD pattern showing broad reflections at $2\theta \approx 24°–31°$. The presence of such crystalline inclusions suggests partial phase separation during synthesis, with local ordering of FAU-type units incorporated into the mesostructure.

The co-existence of ordered mesoporosity and embedded zeolitic nanocrystals reflects a hierarchical architecture that is particularly advantageous for catalysis and adsorption. While the mesopores facilitate efficient mass transport, the zeolitic domains offer active sites with inherent acidity and molecular discrimination. These structural observations provide compelling evidence that MAS(Y) is a composite material, integrating mesostructural regularity with zeolitic functionality.

N_2 adsorption-desorption isotherms and pore size distribution of MAS(Y) material:

Nitrogen adsorption–desorption analysis was performed to quantitatively evaluate the mesoporous architecture of the MAS(Y) material.. The results are presented in Fig 5.

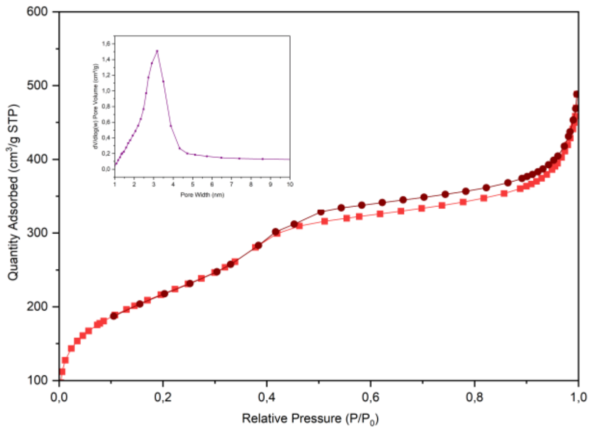

Fig. 5. N_2 adsorption-desorption isotherms and pore size distribution (inset) of MAS(Y) material

As shown in Fig 5, the isotherm profile is characteristic of a type IV curve with an H4-type hysteresis loop according to the IUPAC classification, which is typical for mesoporous solids containing slit-like pores and microporous contributions (X. Liu et al., 2013). At low relative pressures ($P/P_0 < 0.3$), the adsorption uptake increases steadily, indicating monolayer adsorption in micropores. A pronounced inflection at $P/P_0 \approx 0.35$ marks the onset of capillary condensation, signaling the presence of uniform mesopores. The relatively narrow hysteresis loop, with minimal divergence between the adsorption and desorption branches, suggests a narrow pore size distribution and well-defined pore connectivity. The steep increase in adsorption at high relative pressure ($P/P_0 > 0.95$) may be attributed to interparticle condensation, common in ordered mesostructures with open pore networks.

The pore size distribution, derived from the BJH method (inset, Fig 5), reveals a sharp maximum centered at ~3.2 nm, in excellent agreement with the d-spacing observed in the SAXS profile and the inter-channel spacing directly measured by TEM. This congruence across independent characterization techniques confirms the uniformity and regularity of the mesoporous system.

Moreover, the specific surface area, calculated by the BET method, reaches 971 m^2/g, reflecting a high degree of structural porosity. The combination of microporous and mesoporous domains—evidenced by adsorption features at both low and intermediate P/P_0—corroborates the hierarchical nature of the MAS(Y) framework. These features arise from the templated assembly of surfactant micelles and the partial crystallization of zeolite Y domains, as observed in the wide-angle XRD and high-resolution TEM images.

Taken together, the nitrogen physisorption data further substantiate the formation of a highly ordered mesoporous material with integrated zeolitic nanodomains, offering a large accessible surface area and a well-controlled pore architecture suitable for catalytic and sorption-driven applications.

TGA-DTA:

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The MAS(Y) material was synthesized based on zeolite Y seeds derived from activated kaolin. The TGA-DTA diagram (Fig 6) shows several temperature-dependent changes in the material's structure, as follows:

The TGA curve of the MAS(Y) sample shows two regions of weight loss. The first region, occurring at low endothermic temperatures (<200°C), exhibits a weight loss of approximately 24.8%. This effect corresponds to the release of adsorbed water from the surface and pores of the material. The second region shows a slow exothermic effect, with a small total weight loss of only about 7% at temperatures above 900°C.

Therefore, the synthesized MAS(Y) material demonstrates good thermal stability and maintains structural integrity up to 900°C. This is considered a key factor for its potential use as a catalyst, a catalyst support in chemical processes, or as an adsorbent in environmental treatment applications.

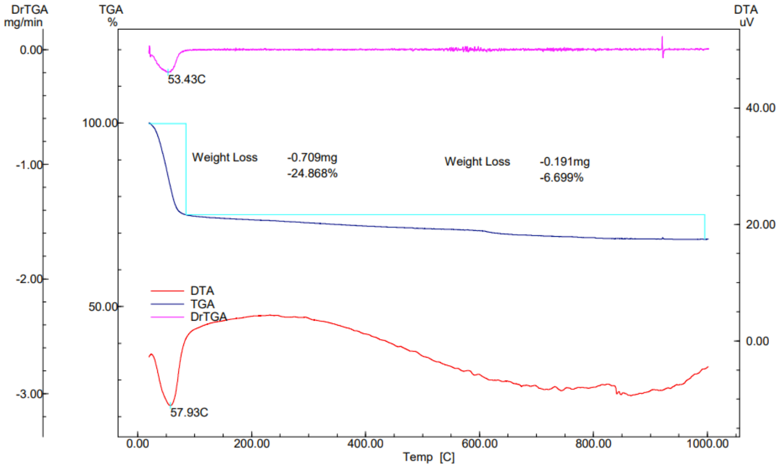
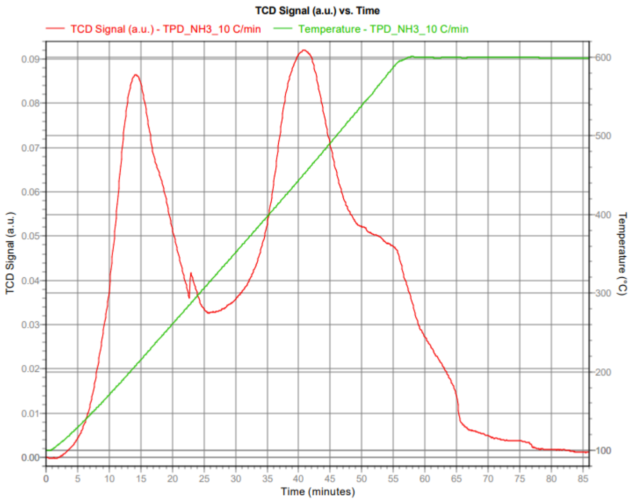


Fig. 6. TGA-DTA analysis of MAS(Y) material

NH_3 temperature programmed desorption (NH3-TPD) analysis:

The acidity profile of the MAS(Y) material was evaluated using temperature-programmed desorption of ammonia (NH_3-TPD), as shown in Fig 7.



Peak Number	Temperature at Maximum (°C)	Volume (mL/g STP)	Peak Concentration (%)
1	207.8	2.14866	-0.31
2	447.0	1.06407	-0.34
3	510.4	5.21029	-0.33

Fig. 7. TPD-NH_3 analysis of MAS(Y) material

The desorption curve reveals three distinct peaks centered at 207.8 °C, 447.0 °C, and 510.4 °C, corresponding to a total desorbed volume of approximately 8.42 mL NH_3/g (STP). These peaks reflect the presence of acid sites with varying strength across the sample surface.

The first desorption peak at 207.8 °C is indicative of weak acid sites. These sites are commonly associated with terminal silanol groups (≡Si–OH) present on the surface of amorphous aluminosilicates and mesoporous walls, consistent with the hierarchical structure of MAS(Y) observed in TEM and SAXS analyses. Their accessibility is enhanced by the high surface area and open mesoporous channels (S_{BET} = 971 m²/g), promoting efficient NH_3 adsorption at lower desorption temperatures.

The second and third peaks, at 447.0 °C and 510.4 °C, are attributed to medium-to-strong Brønsted acid sites. These originate from the framework-incorporated Al^{3+} species located within tetrahedral positions of the zeolite Y nanodomains embedded in the mesoporous matrix (Rahimzadeh et al., 2025). The partial crystallinity and presence of faujasite-type structural units confirmed by XRD and TEM ($2\theta \approx 24°$–$31°$) support the attribution of these high-temperature peaks to protonated bridging hydroxyl groups (Si–OH–Al) within the zeolitic environment (Pinto et al., 2005).

Notably, the high desorption volume observed at 510.4 °C (5.21 mL/g) suggests a significant population of strong acid sites, possibly arising from well-dispersed and partially crystallized FAU-type domains. These acid sites are critical for catalytic applications, providing both strength and stability under elevated temperatures.

In summary, the NH_3-TPD results corroborate the hybrid nature of MAS(Y): the weak acidity originates from the mesoporous framework, while strong acid sites are derived from embedded zeolitic domains. This dual-acidity profile, combined with the hierarchical porosity, renders MAS(Y) a promising material for bifunctional catalysis, where both acid strength and mass transport play essential roles.

Evaluation of the Catalyst Activity in the Co-pyrolysis of PP waste and VGO fraction

To assess the catalytic activity of MAS(Y) in the co-pyrolysis reaction of PP and VGO, experiments were conducted under identical conditions as described in Section 2.4. The liquid products obtained from the co-pyrolysis were analyzed using the GC/MS method. The results are presented in the graph shown in Fig 8.

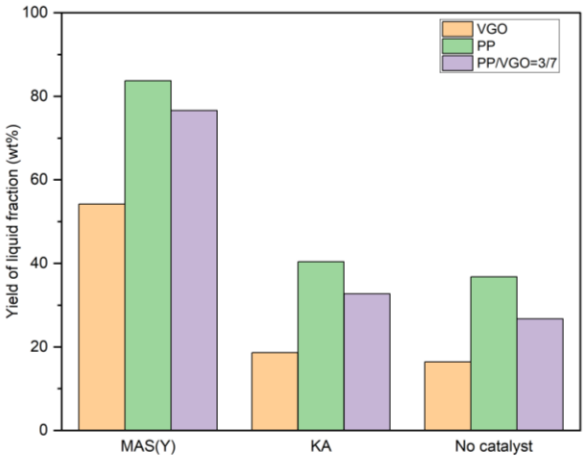


Fig. 8. Catalytic activity on the liquid fraction yield from different feedstocks

The results shown in Fig. 8 indicate that pyrolysis of 100% PP waste feedstock consistently yields the highest amount of liquid product across all three cases using MAS(Y) catalyst, activated Kaolin, and without catalyst. This indicates that PP is a readily pyrolyzable material under the experimental conditions. In contrast, pyrolysis of 100% VGO fraction results in very low liquid yields. In practice, VGO tends to form coke and solid residues on the catalyst surface, which leads to deactivation of catalyst. In addition, the condensable products from VGO pyrolysis tend to have high molecular weights and remain in wax form. To obtain liquid products from VGO, experiments must be conducted at temperatures above 650°C. However, under such harsh conditions, coke formation occurs rapidly, leading to significant catalyst deactivation and very low liquid yields.

For the PP/VGO feedstock mixture with a 3/7 ratio, the liquid yield on the MAS(Y) catalyst reached up to 76%. When waste PP is added as a co-feed, the co-pyrolysis process can proceed under milder conditions (around 500°C), and the liquid product yield increases significantly. According to GC/MS results, an increase in the PP/VGO ratio leads to higher liquid yields. This can be explained by the fact that during PP cracking, a certain amount of hydrogen gas is generated, which assists in the cracking of VGO.

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The presence of hydrogen helps suppress olefin formation and reduces coke deposition, thereby favoring the production of more liquid products.

Individual pyrolysis experiments of PP waste, VGO fraction, and co-pyrolysis of PP/VGO at a ratio of 3/7, both with activated kaolin as catalyst and without catalyst, showed that at 500 °C, the liquid product yields were low in both cases (approximately 40%). This indicates that activated kaolin exhibits negligible catalytic activity for the pyrolysis reaction, and that the non-catalytic process requires higher temperatures to achieve more efficient conversion

However, since the MAS(Y) catalyst is intended for use in the co-pyrolysis of PP and VGO to produce liquid products for various applications, the composition of the liquid product needs to be further analyzed and evaluated.

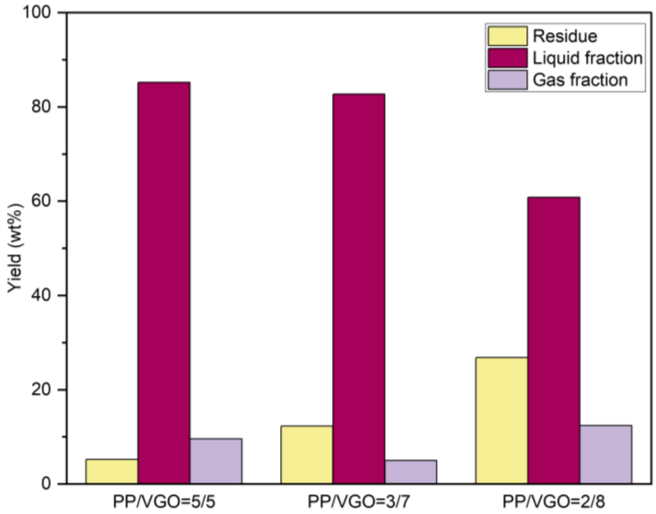

Fig. 9. Graph showing the product composition from PP/VGO feedstocks at different ratios

The liquid products from the co-pyrolysis of PP/VGO feedstocks at different ratios were analyzed using the GC/MS method. The results are presented in the graph shown in Fig. 9.

Fig. 9 shows that the liquid yield increases as the PP/VGO feed ratio rises. This is because PP is inherently easier to crack under the experimental conditions. Additionally, the presence of PP facilitates the pyrolysis of VGO at milder conditions, reducing wax formation and thereby increasing the liquid product yield, which reaches up to 85% at a PP/VGO ratio of 5:5.

However, to better assess the practical applications of the co-pyrolysis products, the liquid products from different PP/VGO ratios were further analyzed.

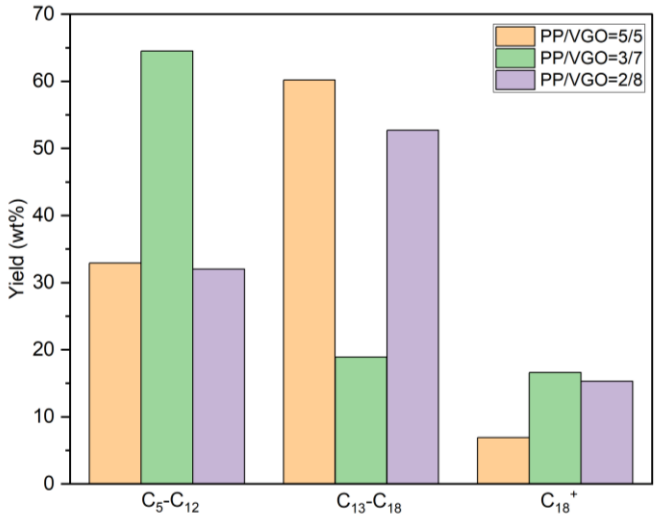

Fig. 10. The composition of liquid fraction from the co-pyrolysis of PP/VGO feedstocks at different ratios

The GC/MS analysis results of the liquid fraction are presented in Fig 10, along with a statistical summary illustrated in the same figure. From the GC/MS data and Fig 10, it can be concluded that to obtain liquid products suitable for gasoline fuel production, the co-pyrolysis process using the HMSU-SY catalyst should be performed at a PP/VGO ratio of 3:7. Meanwhile, a PP/VGO ratio of 5:5 favors the production of diesel-like fuel.

Some images of the liquid products from the co-pyrolysis of PP/VGO feedstocks are shown in Fig. 11.

Fig. 11. Images of the liquid products from the co-pyrolysis of PP/VGO

4. Conclusion

Vietnamese kaolin has been effectively converted from a crystalline kaolinite phase into a highly reactive amorphous form via thermal and acid activation. This transformation significantly enhances its applicability as a precursor for the synthesis of advanced porous materials. In this study, a novel mesostructured aluminosilicate, MAS(Y), was synthesized from activated kaolin using two- step crystillization method that integrates mesophase templating and zeolite nucleation. The resulting material exhibits an ordered hexagonal mesoporous framework, with pore walls incorporating both zeolite Y seeds and partially crystallized zeolite Y nanodomains.

Comprehensive structural analyses including XRD, SAXS, TEM, and N_2 physisorption isotherms confirmed the formation of a hierarchical porous architecture, characterized by uniform mesopores (~3.2 nm), high surface area (971 m²/g), and a dual-level framework combining mesoporosity and zeolitic functionalities. NH_3-TPD results further demonstrated the presence of both weak and strong Brønsted acid sites, confirming the solid acid character of MAS(Y). Notably, the material exhibited excellent thermal stability up to 900 °C, underscoring its structural robustness under severe catalytic conditions.

The catalytic evaluation of MAS(Y) in the co-pyrolysis of waste polypropylene (PP) and vacuum gas oil (VGO) demonstrated high conversion efficiency, with liquid product yields exceeding 80%. Optimal selectivity toward the gasoline-range (C_5–C_{12}) and diesel-range (C_{13}–C_{18}) hydrocarbons was achieved at PP/VGO ratios of 3:7 and 5:5, respectively. These results validate the high potential of MAS(Y) as an effective bifunctional catalyst for upgrading plastic and refinery-derived hydrocarbon streams.

The successful utilization of low-cost Vietnamese kaolin as a raw material for fabricating thermally stable, high-performance solid acid catalysts presents a compelling route toward sustainable catalyst development in petrochemical and waste valorization industries. Future investigations should focus on continuous-flow reactor studies to assess catalytic stability under industrially relevant conditions, including multi-cycle regeneration and deactivation resistance. Furthermore, comprehensive life-cycle and techno-economic assessments will be essential to evaluate the scalability and environmental impact of MAS(Y)-based catalytic processes. Integration of this material into refinery operations or plastic-to-fuel upgrading platforms could offer a cost-effective and circular solution to current waste and energy challenges.

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